

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to:

- **write a parametric representation of an ellipse or circle given its standard form.**
- **write a parametric representation of a parabola using a direct substitution for $x - h$.**
- **use parametric analysis (extrema, axis intercepts) on a parametrized implicit curve.**

Vocabulary & Key Concepts**implicit curve:**

A curve defined by an equation relating x and y that cannot always be written as $y = f(x)$**parametrization:**

A pair of component functions $(x(t), y(t))$ that traces an implicit curve as t varies.**ellipse parametrization:**

 $x(t) = h + a \cos t$, $y(t) = k + b \sin t$, $0 \leq t \leq 2\pi$; from $\cos^2 t + \sin^2 t = 1$**parabola parametrization:**

Set $x(t) = h + t$, $y(t) = k + at^2$, $t \in \mathbb{R}$; vertex at $t = 0$**Hook**

The ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is *not* a function — it fails the vertical line test. But if a particle traces this ellipse over time, we can use parametric tools to find its position, extremes, and intercepts. How do we convert this implicit equation to a parametric form?

Set $x/3 = \cos t$ and $y/2 = \sin t$, giving $x(t) = 3 \cos t$, $y(t) = 2 \sin t$**Section 1 — Ellipse Parametrization**

Setup. Starting from $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, set $\frac{x-h}{a} = \cos t$ and $\frac{y-k}{b} = \sin t$.

Then: $x(t) = h + a \cos t$ and $y(t) = k + b \sin t$ Domain: $0 \leq t \leq 2\pi$

Verify. Substitute back into the ellipse equation and use the identity $\cos^2 t + \sin^2 t = 1$:

$$\frac{(x(t) - h)^2}{a^2} + \frac{(y(t) - k)^2}{b^2} = \frac{(a \cos t)^2}{a^2} + \frac{(b \sin t)^2}{b^2} = \cos^2 t + \sin^2 t = 1 \quad \checkmark$$

Example. Ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Here $h = 0$, $k = 0$, $a = 4$, $b = 2$.

Parametrization: $x(t) = 4 \cos t$ $y(t) = 2 \sin t$ $0 \leq t \leq 2\pi$

Maximum $x(t)$: 4 when $t = 0 \Rightarrow$ Point: (4, 0)

Extrema from the parametrization. Minimum $x(t)$: -4 when $t = \pi \Rightarrow$ Point: (-4, 0)

Maximum $y(t)$: 2 when $t = \pi/2 \Rightarrow$ Point: (0, 2)

Section 2 — Parabola Parametrization

Setup. For $y - k = a(x - h)^2$ (vertical parabola): set $x(t) = h + t$, then $y(t) = k + at^2$, $t \in \mathbb{R}$.

Vertex at $t = 0$: (h, k)

Example. $y = x^2 - 4x + 3$.

Step 1 — Complete the square: $y = (x - 2)^2 - 1$. Vertex: (2, -1).

Step 2 — Parametrize: $x(t) = 2 + t$, $y(t) = t^2 - 1$.

Step 3 — Fill the table:

t	$x(t) = 2 + t$	$y(t) = t^2 - 1$	Point
-2	0	3	(0, 3)
-1	1	0	(1, 0)
0	2	-1	(2, -1)
1	3	0	(3, 0)
2	4	3	(4, 3)

Section 3 — Applying Parametric Tools to Implicit Curves

Ellipse: $x(t) = 3 \cos t$, $y(t) = 5 \sin t$.

- When does the curve cross the y -axis? Set $x(t) = 0$: $\cos t = 0 \Rightarrow t = \pi/2$ and $3\pi/2$
Coordinates at those t -values: (0, 5) and (0, -5)
- When does the curve cross the x -axis? Set $y(t) = 0$: $\sin t = 0 \Rightarrow t = 0$ and π
Coordinates at those t -values: (3, 0) and (-3, 0)
- What is the **rightmost** point? Max of $x(t) = 3 \cos t$ is 3, at $t = 0$. Point: (3, 0)
- What is the **topmost** point? Max of $y(t) = 5 \sin t$ is 5, at $t = \pi/2$. Point: (0, 5)

Teacher Note

Section 3 is the heart of AP Skill 2.A: students must extract information from the parametric representation rather than the Cartesian picture. Emphasize that these intercepts and extrema can be found purely from $x(t)$ and $y(t)$ — no graph required. Common error: students may state the rightmost point is where t is largest, not where $x(t)$ is largest. Use cold-call: “How do you know $t = 0$ gives the maximum of $\cos t$?”